

Which arbitrary choices could change Law & Plank's conclusion, and how to find out

A mizer reimplementation of Law & Plank (2023), Fish and Fisheries

Status: the design document. Results as they come in are in [robustness-results.md](#), which also corrects two claims made here (section 3.1 on the calibration criterion, and the role of `I_0` in section 3.6). It is written against the reimplementation in this repository, and names the code hooks each experiment would need.

It assumes the model set out equation by equation in [the overview](#), and the ecosystem names used there: `eco1` is the 15-species assemblage behind the paper's Figs 2-4, and `eco_r1` - `eco_r3` are three more with the search rate randomised across species.

1. The claim, stated precisely

At matched total yield, setting fishing mortality in proportion to somatic production rate ($F_i = c_P P_i$) leaves rare species roughly where they were after 50 years, whereas a fixed `F` or a fixed exploitation ratio ($F_i = c_P/B P_i/B_i$) puts several of them on an exponential path to extinction.

The paper's stated mechanism is a feedback: because $B_i \sim P_i^\alpha$ with α near 1, $F_i = c_P P_i$ makes fishing mortality roughly proportional to biomass, so it falls away as a stock falls. Fixed `F` has no such feedback, and P_i/B_i changes little as a stock falls, so `BH_P/B` has almost none either.

2. The causal chain, and where it can be attacked

#	Link	What it needs to be true	Attack
i	$B_i \sim P_i^\alpha, \alpha \approx 1$	enough adult mortality that biomass does not pile up at <code>w_max</code> ; a sensible size range over which <code>P</code> is measured	Eq. (A.7); <code>w_f</code> ; the P/B measurement range
ii	$F_i \propto P_i$ therefore $F_i \propto B_i$	mass-specific production roughly density-independent	food-limitation feedback; <code>theta</code>
iii	that feedback is what saves rare species	rare species decline <i>because of fishing</i>	baseline drift; predation release
iv	the other two rules have no comparable feedback	no other density dependence in the model	absence of a stock-recruitment relationship

#	Link	What it needs to be true	Attack
v	"at matched total yield"	the matching criterion is not itself doing the work	terminal-yield calibration; fishing intensity

Links iii, iv and v are the ones the paper does not examine, and they are the ones most likely to matter.

3. Choices that could change the conclusion

3.1 Matching total yield after 50 years (Section 3)

The three constants are calibrated so that all three regimes give the same total yield **in year 50**. By year 50 the fixed- F and BH_P/B ecosystems have been substantially depleted, so their terminal yield is low; matching it forces c_P down and makes BH_P fish *gently throughout*. Matching year-0 yield, or yield integrated over the 50 years, would each give a different c_P — plausibly a much larger one.

This is the single most load-bearing arbitrary choice in the paper, because the whole comparison is "at equal yield" and there are at least four defensible meanings of that phrase. It is not discussed.

Prediction. c_P calibrated on cumulative yield will exceed c_P calibrated on terminal yield, perhaps substantially, and the advantage of BH_P will shrink. Whether it *reverses* is the question.

3.2 No compensatory recruitment (Appendix B)

The model deliberately has no stock-recruitment relationship: recruitment is strictly proportional to adult reproductive output, $RDD = RDI$. The paper presents this as a virtue — density dependence should emerge rather than be imposed — and it is a defensible modelling stance.

But it is also precisely the assumption that makes exponential decline possible. With linear recruitment, a species whose adult mortality exceeds what its lifetime reproductive output can sustain declines geometrically, forever. Add even moderate compensation (a Beverton-Holt R_{max} , i.e. a non-zero `reproduction_level`) and every species acquires a density-dependent brake at low biomass — under *all three* rules. The BH_P feedback would then be supplying protection that the population already has.

Most published multispecies size-spectrum models (including mizer's North Sea model) do impose such a relationship. So this is not an exotic alternative; it is the mainstream one.

The authors are aware that something has to take the place of an imposed stock-recruitment relationship, and nominate a specific mechanism for it — larval competition for plankton (Appendix B, device 1). Section 3.6 shows, by measurement, that at the published parameter values that mechanism is barely operating.

Prediction. As `reproduction_level` rises from 0 towards 0.5, the fixed-`F` collapses turn into new lower equilibria, and the gap between the three rules narrows sharply. If it closes, the paper’s result is a statement about recruitment dynamics as much as about balanced harvesting.

3.3 Which species are declining anyway (Appendix B, Section 3)

The unexploited ecosystem is explicitly *not* at equilibrium: “some species’ biomasses were still changing slowly after 50 years, so the state at this time is best thought of as a quasi-equilibrium”. Sequential assembly with an extinction threshold leaves some species close to exclusion, and some of the “rare” species may have been on their way out before any fishing started.

Every figure compares year 50 *fished* against year 0 *unfished*, so baseline drift and fishing effect are confounded. The fix is trivial — run an unfished control over the same 50 years and compare fished against control rather than against year 0 — and this repository already does it, but the paper does not.

Prediction. Some of the apparent fixed-`F` collapse is drift. The ranking of the three rules probably survives, but the magnitudes will shrink, and possibly by a lot for the rarest species.

3.4 Is it production, or just density dependence?

`BH_P` gives $Y_i = c_P P_i B_i$, and since $P_i \sim B_i$, $Y_i \sim B_i^2$. A rule $F_i = c_B B_i$ gives exactly the same thing without ever mentioning production. If the two behave identically, then the paper’s result is about making fishing mortality density-dependent, and “production” is a convenient proxy for biomass rather than the active ingredient — which matters, because biomass is far easier to estimate than production.

More sharply, the three rules are nearly the same one-parameter family:

$$F_i = c X_i^\theta, \quad \theta = 0 \text{ (fixed } F), \quad \theta \approx 1 \text{ (BH}_P), \quad \theta \lesssim 0 \text{ (BH}_{P/B}).$$

`BHP/B` sits at θ slightly *negative*, because as a stock thins, competition for food eases, `Pi/Bi` rises, and fishing mortality goes *up* — the paper says as much (“The ratio might even increase if low species biomass reduced competition for food”). Sweeping θ continuously turns a three-way categorical comparison into a quantitative question: how much density dependence does a harvest control rule need?

3.5 Adaptive feedback, or just a better allocation?

The paper’s policy claim is specifically about adaptation over time: fishing “operating adaptively to follow species’ production rates over time, contains a feedback that would help to protect species from overfishing in the presence of uncertainty”.

But `BH_P` does two things at once. It *allocates* effort across species at the outset (rare species get low `F` from year 0), and it *updates* that allocation as the ecosystem changes. These can be separated, and the paper does not separate them. The decisive experiment is

a frozen control: set $F_i = c_P P_i(0)$, constant in time. If frozen BH_P does as well as adaptive BH_P , the benefit is the allocation, and the feedback story is decoration.

There is also a second, indirect route the paper mentions only in passing: “we found cases where rare species gained more through release from predation under BH_P than they lost from fishing (results not shown)”. If the protection of rare species is mostly trophic release — because BH_P fishes the abundant large predators hardest — then it depends on the model’s predation structure rather than on the harvest rule’s feedback, and would not transfer to an assemblage with a different trophic arrangement. This is decomposable: compare each rare species’ direct fishing loss against its change in predation mortality.

3.6 The plankton is almost impossible to eat down, so larval competition barely operates

Appendix B nominates larval competition as the first of the three devices that make coexistence possible without an imposed stock-recruitment relationship:

When larval density is low relative to plankton food, the abundance of food allows fast body growth through the vulnerable larval stage, making the risk of death relatively small. Conversely, when larval density is relatively high, growth is slower and the accumulated risk of death in the larval stage becomes greater. This has a strong stabilizing effect [...]

That feedback exists only to the extent that larvae actually deplete their food. They do not. Measured on the assembled 15-species ecosystem `eco1`:

plankton size	$r(w)$ /yr	grazing mortality /yr	resource level n/a
1e-8 g	158	0.04	1.006
1e-6 g	79	0.69	1.004
1e-5 g	56	0.73	1.005
1e-4 g	40	0.87	1.003
1e-3 g	28	1.17	0.994
0.01 g	20	9.73	0.596
0.1 g	14	10.09	0.445
1 g	10	9.91	0.321

A 1 mg larva eats prey between 1e-8 and 1e-4 g. Over that entire range the plankton sits **at its carrying capacity**, despite fifteen fish species grazing it. Only above about 0.01 g, where fish of 0.1 g and up feed, does grazing bite — and there it bites hard, pulling the top of the plankton spectrum down to a third of capacity.

The reason is structural, and follows from Eq. (A.11) in two lines. At steady state, dividing through by $a(x)$ and writing $L = n/a$ and $iota = I/a$:

$$\iota + rL(1 - L) - dL = 0 \quad \Rightarrow \quad rL^2 - (r - d)L - \iota = 0.$$

With $r = 10 w^{-0.15}$, that is 40–220 yr^{-1} across the larval prey range, against a measured grazing mortality of 0.4–0.9 yr^{-1} . Solving for L at $r = 79 \text{ yr}^{-1}$, $d = 0.7 \text{ yr}^{-1}$:

ι (yr^{-1})	1 (published)	0.1	0.01	0
resource level L	1.004	0.992	0.991	0.991

The immigration term is not what protects the plankton. Switching it off entirely moves the resource level from 1.004 to 0.991. What keeps L pinned at 1 is that the plankton regenerate two orders of magnitude faster than they are grazed. (An earlier draft of this section named $I_0 = a_0$ as the decisive choice; that was wrong. It becomes decisive only in combination with a slow plankton — at $r = 0.79 \text{ yr}^{-1}$ the same table reads 1.183, 0.417, 0.183, 0.114.)

So the parameter to reach for is r_0 , and ι matters only once r_0 is already low. Both are set without justification: $r_0 = 10$ is anchored to a cell-division scaling exponent, not to a rate, and I_0 is simply set equal to a_0 with no comment beyond the statement that $I(x)$ “was assumed to scale with body size in the same way as the carrying capacity”.

The effect on growth is direct. Multiplying the whole fish spectrum by 2 and letting the plankton re-equilibrate changes growth rate by:

fish body mass	d log(growth) / d log(fish abundance)
0.001 g (egg)	-0.013
0.01 g	-0.021
0.1 g	-0.093
1 g	-0.271
10 g	-0.524
100 g	-0.908

Doubling the fish nearly halves the growth rate of a 100 g fish and slows an egg by 1.3%. Competition for food is a real and very strong force in this model — but only from about 0.1 g upwards, two orders of magnitude above the egg.

Three consequences.

1. Egg-to-recruit survival is nearly independent of larval density *through growth*.

Since the model also has no satiation ($f = 0$), larval growth rate is essentially a fixed constant, so time spent in the vulnerable stage is fixed, so the growth-mediated part of the larval feedback contributes almost nothing. What density dependence remains at the larval stage comes through *predation* on eggs and larvae by larger fish, which is a different mechanism from the one Appendix B describes.

2. It strengthens the worry in §3.2 rather than relieving it. The model lacks both an imposed stock-recruitment relationship and (at these parameters) the emergent larval mechanism offered in its place. The recruitment brake is therefore weaker than the paper’s own account of the model implies, which makes exponential decline under fixed

F easier and makes the BH_P feedback look more necessary than it would in a model with a working larval feedback.

3. **The absence of satiation cuts the other way, and loses.** With no maximum intake rate, growth is strictly proportional to food, so a 10% drop in food is a 10% drop in growth; in a standard mizer model at a feeding level of 0.6 it would be about 4%. Law & Plank's choice therefore *amplifies* whatever competition exists — and the elasticity at egg size is still only -0.009.

An aside from the same measurement. Averaged over the fifteen species, plankton supplies 100% of intake at egg size and still 98% at 1 kg. By mass flow, every fish in this model is a planktivore. That is a consequence of the box kernel spanning four decades of prey size — the abundant small end dominates the integral — together with the plankton extending up to 1 g. Species are strongly coupled in this model, but through *predation mortality* rather than through competition for fish prey, and a 1 kg fish sits at a trophic level near 2 rather than at the top of a chain. Anything that depends on the length of the food chain (trophic cascades, indirect effects of fishing one species on another) should be expected to be sensitive to the kernel width and to the plankton cutoff of \$4.4.

What to vary. r_0 and ρ (how fast plankton regenerate), I_0/a_0 (the floor that grazing cannot push through), a_0 (standing stock), and λ (which shifts food between small and large plankton). Note that r and I/a control the resource *level* directly, whereas a_0 acts indirectly — raising it raises fish biomass and hence grazing pressure too — so a_0 is closer to a nuisance scale parameter than the other two. Lowering I_0/a_0 by one or two orders of magnitude, or lowering r_0 , should switch the larval feedback on and is the cleanest test of whether the paper's stated coexistence mechanism, once actually operating, changes the comparison between rules.

4. Choices that could change the magnitude

4.1 $w_f = 400$ g, a single knife edge for every species

The paper flags this as a simplification (“we picked a single illustrative pattern of fishing [...] there are clearly many other possibilities”), but does not test it, and it is more consequential than it looks. With $w_{mat} = w_{max}/10$ for every species:

- a 1 kg species matures at 100 g and is fished only as an adult;
- a 40 kg species matures at 4 kg and is fished for a decade before it breeds.

So a fixed w_f combined with a fixed w_{mat}/w_{max} ratio *systematically* subjects large species to recruitment overfishing and shields small ones. Some of “large species are vulnerable, small ones are not” is built in by this choice rather than discovered. Species with $w_{max} < 400$ g are not fished at all.

4.2 Measuring P and B over the harvested range only

Justified on data-availability grounds (“reliable information is most likely to be available over this range”), but it makes P_i a strange quantity: it excludes juvenile production,

which is most of a species' somatic production, and for a species whose $w_{\max}$ is only a little above w_f it is dominated by the boundary influx term at w_f . That makes P_i — and hence F_i under BH_P — very sensitive to where w_f falls in a species' life cycle.

4.3 The interaction matrix (Appendix C)

$\theta_{ii} = 0.5$, $\theta_{ij} = 0.2$, flat, chosen to give diagonal dominance because that “is known to promote coexistence”. Two issues. First, the 2.5:1 ratio is not measured, and coexistence — hence which species are rare — depends on it. Second, strong cannibalism is itself a density-dependent brake, so it may be supplying some of the stability attributed to BH_P . Real matrices (Blanchard et al. 2014; Spence et al. 2021) are heterogeneous, and with a flat off-diagonal all 15 species are ecologically interchangeable apart from $w_{\max}$.

4.4 Plankton capped at 1 g

(See also §3.6, which is about the same spectrum but a different property of it.)

Admitted to be artificial, “to compensate for the absence of multicellular zooplankton”. With a feeding kernel spanning prey from 1/100000 to 1/10 of body mass, every fish above about 10 g must get part of its diet from fish. Raising the cap weakens fish–fish coupling and therefore weakens both the trophic cascades and the predation-release effect of §3.5.

4.5 Fishing intensity

The paper already notes that adaptive fishing “worked better at lower, than at higher fishing intensities”. The baseline is doubled from 0.1 to 0.2/yr between Fig. 3 and Fig. 6 and the BH_P time series become visibly less faithful. Since the conclusion is intensity-dependent, reporting it at one or two intensities is weaker than reporting the whole trade-off curve (§5.1).

4.6 Numerical resolution

$dx = 0.1$ in log body mass. The production integral of Eq. (2.4), with its boundary terms, is only first-order accurate: in this implementation the discretisation error is about 10% at $dx = 0.1$, falling to 2.5% at $dx = 0.025$ (`tests/test_model.R`). Under BH_P , F_i is *proportional* to that quantity, and the error is not the same for every species, so it acts as a species-dependent bias in the harvest rule. Worth a grid-convergence check on the results, not just on the rates.

5. Three changes that dissolve arbitrariness rather than testing it

5.1 Report the trade-off frontier, not a single calibrated point

Instead of picking one c_P and comparing, sweep the intensity constant for each rule over two orders of magnitude and plot total yield against a biodiversity metric. Each rule becomes a curve. If BH_P 's curve dominates the other two over the whole range, the conclusion is robust to *every* choice of yield-matching criterion at once, and §3.1

evaporates. If the curves cross, the conclusion holds only in some yield range, and it matters which.

This is strictly more informative than the paper's design and costs only about eight runs per rule.

5.2 Compare against an unfished control, not against year 0

Removes the confound of §3.3 at negligible cost.

5.3 Replicate across assemblages

The paper has one ecosystem for Figs 2–4 and three for Figs 5–6. Between-assemblage variation in assembly-based models is typically large, and with $n = 4$ a consistent ranking is suggestive, not established. Ten to twenty independent assemblages would let the frontier be reported with a spread.

6. Probably safe

Kernel shape at fixed mean predator:prey mass ratio; the diffusion term; egg mass; eps_R ; the exact number of species; rho_L and x_L ; the immigration rate I_0 (it only matters where plankton are grazed to near zero). These are worth a single one-at-a-time check late on, not a campaign.

7. The plan

Metrics, computed at year 50 for every run, always relative to the unfished control rather than to year 0:

- $\min_i (B_i/B_i^{\text{ctrl}})$ — the worst-hit species, the paper's implicit measure;
- number of species below 10% and below 1% of control;
- RMS of $\log(B_i/B_i^{\text{ctrl}})$ — overall distortion of the assemblage;
- change in Shannon diversity of the biomass vector;
- number of species below the assembly extinction threshold;
- total yield, and cumulative yield over the 50 years.

Phase 1 — dissolve the calibration question (highest value, lowest cost)

1. **Yield-biodiversity frontier.** For each of the three rules, eight intensity multipliers spanning 0.1x to 10x the calibrated constant; 50-year runs; plot yield against each metric. Answers §3.1 and §4.5 together.
2. **Unfished control.** Already implemented; fold into every comparison (§3.3).
3. **Alternative calibration criteria.** Recalibrate on year-0 yield and on cumulative yield, and report how much c_P moves.

Cost: about 24 runs per ecosystem at $dt = 0.01$, roughly half an hour each.

Phase 2 — is the mechanism what is claimed?

4. **Frozen-allocation controls.** $F_i = c_P P_i(\theta)$ held constant, and $F_i = c_P/B_i(\theta)/B_i(0)$ held constant. Separates allocation from feedback (§3.5).
5. $F_i = c_B B_i$. Tests whether production is doing any work (§3.4).
6. **The theta sweep.** $F_i = c B_i^\theta$ for θ in $[-0.5, 1.5]$, calibrated to a common yield. Places all rules on one axis.
7. **Predation-release decomposition.** For each rare species, track fishing mortality and predation mortality separately over the 50 years, and report how much of the biomass change under BH_P is attributable to each.

Cost: about 20 runs per ecosystem.

Phase 3 — model assumptions

8. **Reproduction.** `reproduction_level` in $\{0, 0.1, 0.25, 0.5, 0.75\}$, refitting the steady state each time, then the full three-rule comparison (§3.2). This is the most likely single result-changer after §3.1. 8b. **Resource level.** Sweep I_0/a_0 in $\{1, 0.1, 0.01, 0\}$ per year and r_0 in $\{1, 3, 10, 30\} g^{(1-\rho)}/yr$, reassembling each time; measure the resource level across plankton size and the growth elasticity at egg size for each, then run the three-rule comparison on the variants where larval competition is actually operating (§3.6). Pair with item 8: together these two items determine how much recruitment compensation the model has, by either route.
9. **Eq. (A.7).** Repeat the headline comparison under `mu_b_form = "product"`, and sweep `mu_b0` and `xi`. Reports how α in $B \sim P^\alpha$ moves, since the Fig. 4 argument is stated in terms of α (see `docs/mu_b.md`).
10. **w_f.** 100, 200, 400, 800 g; plus a species-relative entry size `w_f = w_mat` and `w_f = w_max/4`, which removes the systematic penalty on large species (§4.1).
11. **Measurement range for P and B.** Harvested range (as published) versus whole life cycle (§4.2), with `w_f` held at 400 g.
12. **theta.** Diagonal:off-diagonal ratios of 1:1, 2.5:1 (published), 5:1; plus one heterogeneous matrix (§4.3).
13. **Plankton cap.** 1 g (published) versus 10 g (§4.4).
14. **Grid.** `dx` of 0.1, 0.05, 0.025 on the headline comparison (§4.6).

Cost: each variant needs its own assembly, since changing the model changes which species coexist — roughly 15 minutes of assembly plus half an hour of runs per variant.

Phase 4 — how the rule would actually be implemented

5. **Update interval.** F_i recomputed every 1, 3, 5 years from the state at that moment rather than continuously. Continuous perfect feedback is a strong idealisation for a policy claim.
6. **Observation error.** $F_i = c_P P_i e^{\eta_i(t)}$ with log-normal noise, and with a persistent per-species bias. The paper tests a *fixed* per-species intensity factor $z' \sim U(0.5, 1.5)$, which

is bias without noise and without dynamics; this is the harder test of the robustness-to-uncertainty claim.

Phase 5 — replication

.7. Repeat Phases 1 and 2 across 10–20 independently assembled ecosystems and report distributions rather than exemplars (§5.3).

Ordering

Phase 1 first: if the frontier shows BH_P dominating everywhere, §3.1 disappears and the rest of the campaign is about *why* rather than *whether*. If the frontiers cross, everything downstream has to be conditioned on yield level. Phase 2 next, because it is cheap and tests the stated mechanism directly. Phase 3 item 8 (recruitment) is the most likely to overturn the headline and should not be left to the end.

8. What each outcome would mean

Finding	Reading
BH_P frontier dominates at all yields, in most assemblages	The conclusion is robust and stronger than the paper shows
Frontiers cross	The conclusion is a statement about low-intensity fishing; say so
Frozen BH_P ~ adaptive BH_P	The benefit is allocation, not feedback; the policy claim needs rewording
$F \propto B \sim F \propto P$	Production is a proxy; use biomass, which is far easier to estimate
Gap closes as reproduction_level rises	The result is contingent on the absence of compensatory recruitment
Gap closes with species-relative w_f	Much of the effect came from a fixed w_f penalising large species
Most of the rare-species gain is predation release	The mechanism is trophic, not the harvest rule’s feedback
Gap closes once larval competition is switched on (lower I_0/a_0 or r_0)	The result depends on the recruitment brake being absent, and the paper’s own stated coexistence mechanism was not operating at its published parameter values

9. Code hooks needed

Already parameterised: mu_b_form, theta_ii / theta_ij (LP_FISH), the plankton cap (LP_PLANKTON\$w_max), dx (LP_NUMERICS), w_f (lp_set_fishing), the per-species intensity factor zf.

To add:

- ~~new rules in `lp_F()`~~ — done: "BHB", "power" and "frozen", all sharing the single form $F_i = c z'_i g_i$;
- ~~a measurement range for `P` and `B` independent of the harvested range~~ — done, as `other_params$lp_jm` alongside `lp_jf`;
- ~~an update interval and an observation-error model~~ — done, as `lp_harvest_periodic()`: the projection runs in segments, and at each segment boundary the state is observed (optionally with log-normal error), F_i is frozen at that value, and the segment is projected;
- ~~`reproduction_level(params) <- ...`~~ — done, and no re-relaxation is needed: `setBevertonHolt()` rescales `erepro` so the steady state is preserved exactly, which is what makes Phase 3 item 8 a clean experiment;
- species-relative `w_f`, which makes the selectivity a full species-by-size array rather than one shared mask;
- a mortality-decomposition recorder for item 7, tracking `f_mort` and `pred_mort` separately per species over time;
- a resource-level diagnostic — `n_pp / resource_capacity` by size, the growth elasticity to fish abundance, and the split of egg-to-recruit survival into its growth-mediated and predation-mediated parts (§3.6). The plankton parameters themselves are already exposed in `LP_PLANKTON`.

Page <https://gustavdelius.github.io/balanced-harvest/robustness.html> of the site accompanying the mizer reimplementa-
tion of Law & Plank (2023). Generated by `make_pdfs.py`.